

**Motorola 68000, Dual Core, Core 2
duo, Core i3, Core i5, Core i7**

Motorola 68000

- CISC processor
- sixteen 32-bit registers
 - eight general purpose *data* registers
 - eight general purpose *address* registers
- User/supervisor space
- 64-pin package
- Clock 8MHz, 12.5 MHz

What are dual core processors?

- Refers to a CPU that includes two complete execution cores per physical computer system.
- Combines two processors and their cache and cache controllers onto a single chip.
- The two processors can be either located on the same motherboard or on separate boards.
- The two identical processors are manufactured so they reside side by side on the same die.

Requirements of a dual core system

- The OS must be able to recognize multithreading.
- Software must have simultaneous multithreading technology(SMT) written into its code.
- A new motherboard is not necessary as they can be used in existing board if that features the correct socket.

Advantages

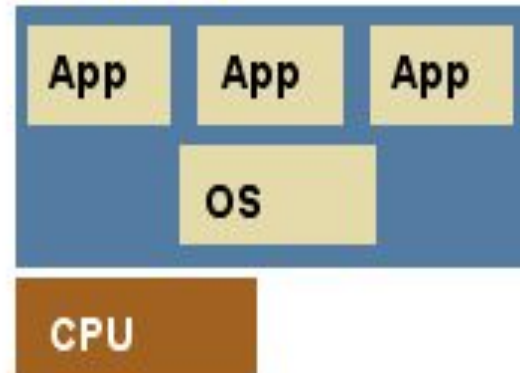
- Dual core can boost system's multitasking computing power.
- It provides two separate complete execution cores.
- As each core has its own cache the processor has sufficient resources to handle intensive task in parallel.
- Even without a multithread enabled application dual core processor are beneficial if you are running an OS that supports thread level parallelism or TLP.

DUAL CORE PROCESSOR

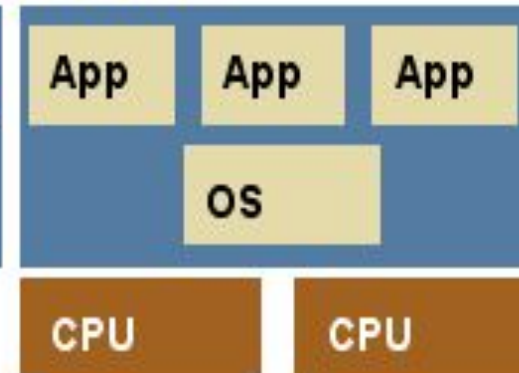
A dual core processor is a CPU with two separate cores on the same die, each with its own cache. It's the equivalent of getting two microprocessors in one.

From Computer Desktop Encyclopedia
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SINGLE CORE



DUAL CORE



SOME DETAILS ABOUT DUAL CORE

Dual core chip is that tasks can be carried out in parallel streams, decreasing processing time. This is referred to as *thread-level parallelism (TLP)*.

TLP is also possible on motherboards that can accommodate two separate CPU dies. When TLP is accomplished in a single CPU through dual core technology, it is called *chip-level multiprocessing (CLM)*.

In dual core CPUs, each microprocessor generally has its own on-board cache, known as *Level 1 (L1)* cache. L1 cache significantly improves system performance, because it is much faster to access on-chip cache than to use random access memory (RAM). L1 cache is accessed at microprocessor speeds.

Dual core chips also commonly feature secondary shared cache on the CPU, known as *Level 2 (L2)* cache. Motherboards may also have a cache chip designated as *Level 3 (L3)* cache. While faster than RAM, L3 cache is slower than cache built into the dual core chip.

DIFFERENT FROM SINGLECORE PROCESSOR

In a single-core or traditional processor the CPU is fed strings of instructions it must order, execute, then selectively store in its cache for quick retrieval. When data outside the cache is required, it is retrieved through the system bus from random access memory (RAM) or from storage devices. Accessing these slows down performance to the maximum speed the bus, RAM or storage device will allow, which is far slower than the speed of the CPU. The situation is compounded when multi-tasking. In this case the processor must switch back and forth between two or more sets of data streams and programs. CPU resources are depleted and performance suffers.

In a dual core processor each core handles incoming data strings simultaneously to improve efficiency. Just as two heads are better than one, so are two hands. Now when one is executing the other can be accessing the system bus or executing its own code. Adding to this favorable scenario, both AMD and Intel's dual-core flagships are 64-bit.

ITS UTILITIES

To utilize a dual core processor, the operating system must be able to recognize multi-threading and the software must have simultaneous multi-threading technology (SMT) written into its code. SMT enables *parallel* multi-threading wherein the cores are served multi-threaded instructions in parallel. Without SMT the software will only recognize one core. Adobe Photoshop is an example of SMT-aware software. SMT is also used with multi-processor systems common to servers.



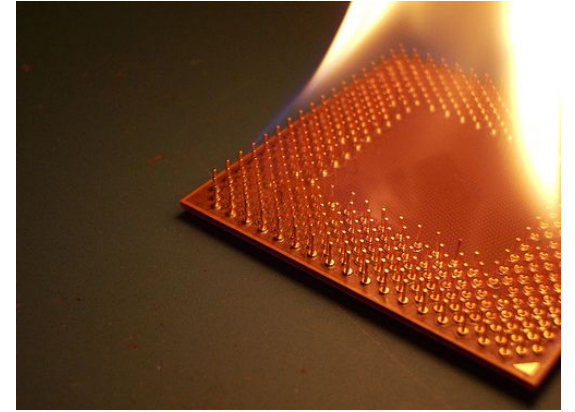
DIFFERENT FROM MULTI PROCESSOR SYSTEM

A dual core processor is different from a multi-processor system. In the latter there are two separate CPUs with their own resources. In the former, resources are shared and the cores reside on the same chip. A multi-processor system is faster than a system with a dual core processor, while a dual core system is faster than a single-core system, all else being equal.

An attractive value of dual core processors is that they do not require a new motherboard, but can be used in existing boards that feature the correct socket. For the average user the difference in performance will be most noticeable in multi-tasking until more software is SMT aware. Servers running multiple dual core processors will see an appreciable increase in performance.

Intel Core 2 Duo

- Intel Core 2 Duo
 - A dual-core CPU
 - ISA with SIMD Extension
 - Intel Core microarchitecture
 - Memory Hierarchy System

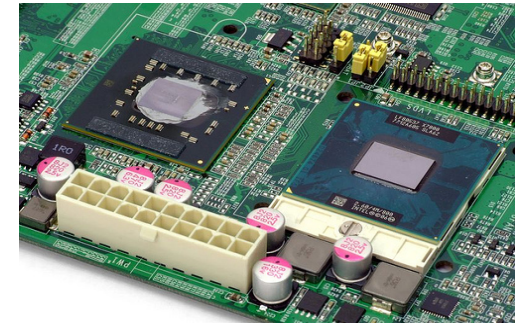


Core 2 Duo



The Intel Core 2 Duo (also known as Core2 Duo) processor is a **64 bit** dual core processor. This means **two processor cores** work inside a Core 2 Duo in **parallel**.

The Core 2 Duo, which was introduced on July 27 2006, is the direct successor of the Core Duo. Each core is based on the Pentium M micro architecture. Compared with the old Net burst architecture of the Pentium 4, the cores in the Core 2 Duo have **shorter pipelines**. As a result, the maximum clock rate is lower but the performance per clock is significantly higher. Thus a Pentium 4M with the same clock rate is up to 40% slower.



Key Features

- ✓ 64 bit support
- ✓ Dual core processor with shared level 2 cache
- ✓ Execute Disable Bit
- ✓ Partially Intel Virtualization Technology (VT)
- ✓ Socket M (starting from Santa Rosa socket P)
- ✓ 291 million transistors

Features and benefits

With an Intel® Core™2 Duo processor you will get performance-rich technologies, including up to 6MB of shared L2 cache, up to 1333 MHz Front Side Bus, plus these additional Intel® technologies built in:

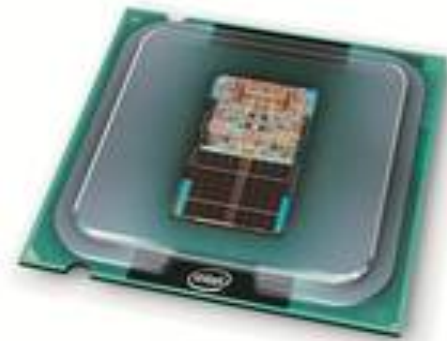
□ Level-1 and Level-2 caches are the cache memories in a computer Much faster than RAM L1 is built on the microprocessor chip itself. L2 is a separate chip L2 cache is much larger than L1 cache

PERFORMANCE

The performance of the Core 2 Duo processor in comparison with its direct competitor, the AMD Turion 64 X2, is very good. The Core 2 Duo beats the AMD Turion 64 X2 with the same clock rate in virtually all applications (on average by 15%). The energy consumption is similar in both processors. In comparison with the predecessor Core Duo, the Core 2 Duo is about 10% faster and consumes a bit more energy

Intel® multi-core processing

provides greater multitasking performance by combining two independent processor cores in one physical package



The Core 2 Duo processors are produced in 65 nm (and later in 45nm), contain 14 stages pipelines and 2-4 MB level 2 cache (depending on the model). The features of the Core 2 Duo are:

x86 architecture

The Intel Core 2 Duo uses the x86 instruction set, which was introduced in 1978 with the 8086/8088 processor. Furthermore it supports the multimedia extension MMX, SSE2, SSE3 and SSE4.

Dual core technology

Two processor cores run with the same frequency in the same processor building block and share the level 2 cache as well as the front side bus (FSB).

Execute Disable Bit

Prevents security problems through buffer overflows, if the operation system supports it and if it is activated.

Wide Dynamic Execution *

Every core can execute four complete commands simultaneously.

Smart-Memory-Access *

Shorter idle times, improved data transfer and faster out-of-order command execution lead to better usage of the pipeline and as a result to higher performance.

Advanced-Smart-Cache *

Like the Core Duo, the Core 2 Duo has shared level 2 cache and each core receives the same amount of cache. Intel doubled the bandwidth to the level 1 cache though.

Advanced-Digital-Media-Boost *

One 128-Bit SSE command is now output per clock cycle

Virtualization technology (VT)

The Intel VT offers hardware support for virtual systems on one computer (use of several isolated operation systems at the same time e.g. through Xen or VMWare).

Beware, not all models support VT (especially the cheaper ones don't).

64 bit support *

Support of 64 bit wide words in the CPU. This means the processor can handle 64 bit data packets. The Intel Core 2 Duo supports the AMD64 extension (licensed), through which 32 and 64 bit programs can run on the CPU (if a 64 bit operation system is used).

Theoretically, a 64 bit processor can access more than 4 GB of memory, but this is usually limited by the chip set used.

*** This feature was newly introduced with the Core 2 Duo.**

Concept of GENERATION

- The term Generation is applied to Intel processor to mean new and significant developments in processor architecture or function.
- And all Intel team is ultimately the one deciding the names 2nd, 3rd, 4th and so on generation.

4th Generation Intel Core Processor & Advantages

- 4th Generation Intel Core processor, more advanced, faster and new line of low-power processor, have arrived in 2013, aims to set a new standard for mobile computing experience and innovative Ultrabook, and tablets designs



Intel i3 processor.

- The only question I have is if there's any advantage to using a Core i3 over a Core2Duo.
- The Core i3 is a 3.2 GHz dual core chip, and it's quite a big improvement over the Core2Duo. It's a good 700 MHz faster, much faster bus with significantly faster RAM.
- The first Core i3 processors were launched on January 7, 2010.

feature of i3

- The Core i3 is a 3.2 GHz dual core chip, and it's quite a big improvement over the Core2Duo.
- It's a good 700 MHz faster, much faster bus with significantly faster RAM.
- Uses less heat and energy than earlier processors, which is always a good thing in this day and age.

Available i3 processor.

Processor Number	Cache	Clock Speed	#of cores #of threads	Max power	Graphics
I3-4000M	3.0 MB	2.40 GHz	2/4	37	Intel HD Graphics 4600
I3-4005U	3.0 MB	1.70 GHz	2/4	15	Intel HD Graphics 4400
I3-4100E	3.0 MB	2.40 GHz	2/4	37	Intel HD Graphics 4600
I3-4102E	3.0 MB	1.60 GHz	2/4	24	Intel HD Graphics 4600

i3 over core 2 duo.

		
Processor	Intel® Core™ 2 Duo P8400 Processor	Intel® Core™ i3-350M Processor
Codename	Penryn	Arrandale
Manufacturing Process	45 nm	32 nm (CPU), 45 nm (Gfx)
# of Transistors	410 Million	382 + 177 Million
# of Cores	2	2
# of Threads	2	4
Micro-architecture	Core micro-architecture	Core micro-architecture
Clock Speed	2,26 GHz	2,26 GHz
L2 Cache	3 MByte	3 MByte
Intel® Turbo Boost Technology	No	No
Intel® Hyper-Threading Technology	No	Yes
Intel® Virtualization Technology (VT-x)	Yes	Yes
Enhanced Intel® Speedstep Technology	Yes	Yes
Instruction Set	64-bit	64-bit
Integrated Memory Controller	No	Yes
Memory Support	defined through chipset	DDR3-800/1066
Max. Memory Size	defined through chipset	8 GByte
# of memory channels	defined through chipset	2
Integrated Graphics	defined through chipset	Yes

Intel i5 processor.

- I5 provides the opportunity to the users to use the system with multi tasking.
- A big feature of the I5 processors is that they have ability to run two multitasking processors together that are generally called as dual processors and can increase the working performance of the system efficiently.
- Turbo boost technology of I5 processors is the key beneficial feature of the I5 processors that allow the users to do their regular and important working with the help of heavy applications.

Features of i5 processor.

- I5 processors have ability to work with integrated memory and can enhance the performance of the applications. The increase the memory up to 1333 MHz
- as I5 processors have high speed performing rate so they are able to perform at the maximum CPU rate of 3.6 GHz
- turbo technology is present in the device that boost up the working speed of the computational systems
- It provides the 64 bit architecture for the users for the reliable working.
- Micro architecture for the I5 processors was presented by the Nehalem and these processors have a cache rate up to 8 MB.

Available i5 processor.

Processor Number	Cache	Clock Speed	#of cores #of threads	Max power	Graphics
I5-4200H	3.0 MB	2.80 GHz	2/4	47	Intel HD Graphics 4600
I5-4440	6.0 MB	3.10 GHz	4/4	84	Intel HD Graphics 4600
I5-4440S	6.0 MB	2.80 GHz	4/4	65	Intel HD Graphics 4600
I5-4400E	3.0 MB	2.70 GHz	2/4	37	Intel HD Graphics 4600

i5 over i3.

- This is the mid-size processor of this bunch, recommended for those who demand a little speed, but not quite enough where the user will be able to run intensive applications.
- As with the Core i3 processor, this comes with 2-4 cores, the main difference is that it has a higher clock speed than the Core i3.
- This is also a heat and energy efficient processor, but it does seem to be better at this particular job than the Core i3.
- The number of threads used in this is no different than the Core i3 with 2-4 threads , it also uses hyper threading technology for a boost in performance.
- The cache of the Core i5 is bigger than the Core i3.
- The Core i5 is in turbo mode is made available, this provides users with the opportunity to turn off a core if it's not being utilized.

Intel i7 processor.

- Intel developed different types processors for the users for the sake of faster working as compared to the previous versions.
- After the development of I3 and I5 processor Intel introduces a new version i.e. I7 processor that is comparatively faster than the old ones.
- Basically I7 processors were designed to meet the challenges of the intelligent and the faster working performance of the computer system.
- They were designed in 2008 and were announced in 2009 just after the I5 processors.

Feature of i7 processor.

- With the help of I7 processors users can enjoy the high speed working with the additional feature of the multitasking i.e. using two different documents or the files at the same time.
- Dual core technology is also a part of I7 processors that provide reliable and high data rate working performance to the customers.
- A big feature of the I7 processors is the Turbo Boost technology. This technology provides the high performance to the system to the users to overcome the work load of different applications on the system and maintain the speed of the system.
- Like I5 and I3 processors I7 also have a feature of Hyper threading technology enhances the activity and the speed of the system by managing the multitasking, work load and different types of heavier applications of the users.
- I7 processors are so advanced to deals with the integrated memory of e system and have ability to increase the memory up to 1066 Mbits and provide the working speed of 25.6 GB/sec.

Available i7 processor.

Processor number	Cache	Clock speed	#of cores #of threads	Max power	Graphics
I7-4960HQ	6.0 MB	2.60 GHz	4/8	47	Intel Iris Pro graphics 5200
I7-4600M	4.0 MB	2.90 GHz	2/4	37	Intel Iris Pro graphics 4600
I7-4600U	4.0 MB	2.10 GHz	2/4	15	Intel Iris Pro graphics 4400
I7-4610Y	4.0 MB	1.70 GHz	2/4		Intel Iris Pro graphics 4200

i7 over i5.

- This is for the users that demand power,
- The cache on this one is for the users that demand more cache memory.
- The cache on this one is 4-8 MB.
- This processor comes with 8 threads, definitely enough to get the job done quickly, maybe even at the speed of light. And yes it also utilizes hyperthreading technology.
- You will have four cores to take advantage of with this particular series.
- And just like the other ones in this Intel series of processors, it is more energy efficient and produces less heat.